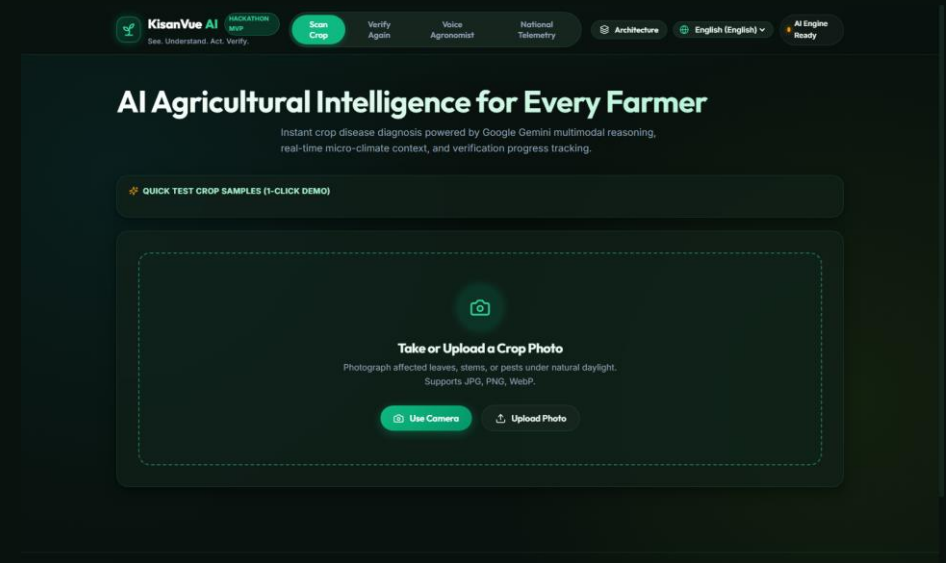


GOOGLE CLOUD · BUILD WITH AI · TRACK 4

KISANVUE AI

See. Understand. Act. Verify.

AI-Powered Agricultural Intelligence for India
Turning multimodal imagery, satellite signals, and soil dynamics into verified farmer decisions.



Multimodal AI

Google Gemini Reasoning

Earth Observation

Sentinel-2 NDVI & NDWI

Soil & Weather

SoilGrids & Open-Meteo

Closed-Loop

Verify-Again Engine

LIVE PLATFORM: <https://kisanvue-ai.vercel.app/> | BACKEND: FastAPI + Render | OPEN SOURCE

The Challenge: Farm Decisions Are Still Fragmented

Small and marginal farmers need timely, localized guidance — but vital agricultural intelligence remains isolated in silos.

Weather Forecasts

Disconnected Micro-Climates

Rainfall or humidity alerts sit on general apps without linking to active fungal incubation windows or spraying timing.

Satellite Imagery

Inaccessible Earth Obs

Sentinel-2 NDVI & NDWI data exists in complex GIS dashboards far removed from daily smallholder comprehension.

Soil Test Reports

Static Physical Cards

Soil organic carbon (SOC) and texture (clay/sand) metrics remain in drawer records rather than guiding immediate inputs.

Crop Health Advice

Fragmented Field Diagnosis

Farmers receive isolated visual symptom guesses without validating against local climate, soil suitability, or past advisories.

CORE PROBLEM STATEMENT:

"How do we turn fragmented agricultural data into one actionable, verified farmer decision?"

Our Solution: One Unified Intelligence Layer for the Farm

KisanVue AI fuses multimodal image reasoning with real-time field context into a single actionable advisory loop.

1. Google Gemini Multimodal AI

Visual pathology classification, lesion identification, and contextual agronomic reasoning.

2. Multi-Source Context Fusion

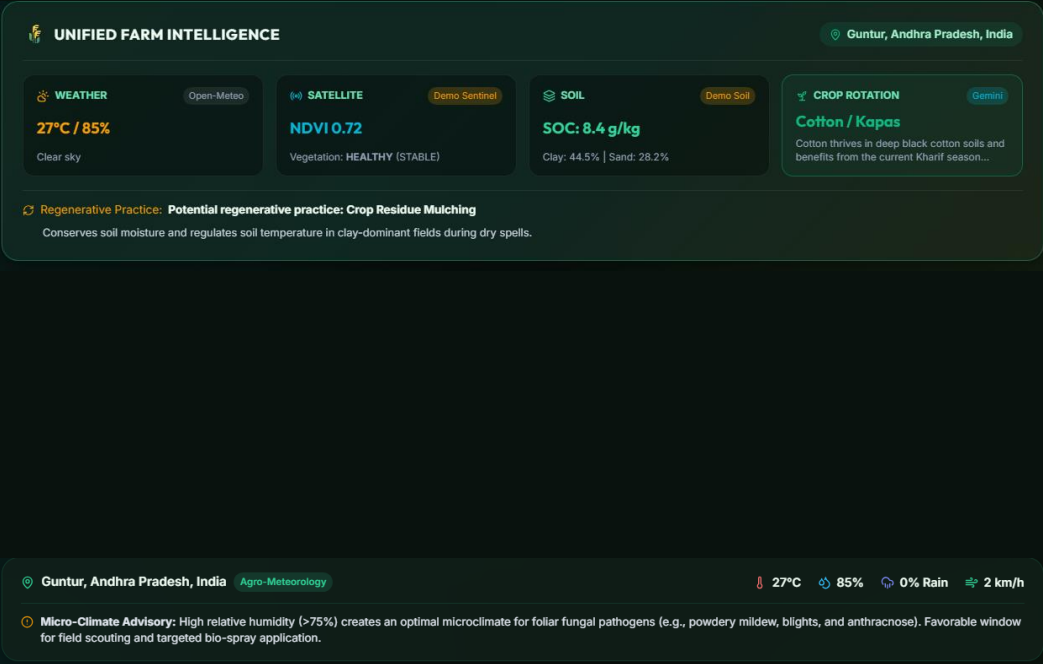
Integrates Open-Meteo weather, Sentinel-2 vegetation indices, and SoilGrids soil properties.

3. Regenerative Crop Planning

Soil- and weather-grounded crop rotations and regenerative organic soil practices.

4. Closed-Loop 'Verify Again'

Post-treatment comparative visual assessment tracking recovery progress over time.



The Core Workflow: From Field Observation to Verified Action

1. SCAN

Farmer uploads crop image or captures live photo under field daylight.

Supports leaf, stem, or vector infestation imagery.

2. UNDERSTAND

Google Gemini performs multimodal reasoning on foliar lesions.

AI-assisted visual assessment without unsupported claims.

3. CONTEXT

Fuses real-time Weather, Sentinel-2 NDVI, and SoilGrids.

Grounds visual diagnosis in active environmental reality.

4. RECOMMEND

Generates succession crops & regenerative practices.

Legume rotation, bio-mulching, and organic soil health.

5. ACT

Delivers localized, multilingual IPM bio-advisory.

Neem oil, sticky traps, balanced fertigation (Telugu/Hindi).

6. VERIFY

Farmer returns 5–7 days later with follow-up photo.

AI-assisted visual comparison computes recovery score.

Multimodal Crop Intelligence: Gemini-Powered Visual Reasoning

Google Gemini turns raw foliar imagery into structured agronomic reasoning with

AI-Assisted Visual Assessment

Clearly designated as visual advisory rather than laboratory-grade pathology, avoiding false claims and maintaining farmer trust.

Observed Foliar Indicators

Detects chlorosis, leaf curling, necrosis, fungal sporulation, and insect vectors (e.g., Bemisia tabaci).

Context-Aware Severity Scoring

Assigns calibrated risk levels (LOW / MEDIUM / HIGH) synthesized with local humidity and temperature windows.

Safe Actionable Advisory

Emphasizes Integrated Pest Management (IPM), bio-controls, and clear escalation thresholds to Krishi Vigyan Kendras (KVK).



GEMINI INFERENCE

Condition: Chilli Leaf Curl
Risk: HIGH (Vector active)
Confidence: 89%
Vector: Bemisia tabaci
Advisory: Neem oil bio-spray

Guntur, Andhra Pradesh, India Agro-Meteorology

27°C 85% 0% Rain 2 km/h

Micro-Climate Advisory: High relative humidity (>75%) creates an optimal microclimate for foliar fungal pathogens (e.g., powdery mildew, blights, and anthracnose). Favorable window for field scouting and targeted bio-spray application.

TRANSPARENCY STANDARD:

Every diagnostic card displays 'AI-assisted advisory' to keep expectations realistic and scientifically responsible.

Environmental Intelligence: Earth Observation & Field Context

Diagnosis becomes significantly stronger when visual symptoms are cross-referenced with satellite, soil, and atmospheric data.

SATELLITE INTELLIGENCE

Sentinel-2 Multi-Spectral

- NDVI: 0.72 — Healthy vegetative density
- NDWI: 0.28 — Adequate canopy water index
- Trend: STABLE — Multi-temporal consistency
- Provider: — Live API + Domain Fallback

SOIL INTELLIGENCE

SoilGrids Global Dataset

- SOC: 8.4 g/kg — Moderate organic carbon
- Clay: 44.5% — Clay-dominant moisture retention
- Sand: 28.2% — Balanced structural drainage
- Texture: — Clay Loam field profile

WEATHER INTELLIGENCE

Open-Meteo Agro-Climate

- Temp: 27°C — Moderate field temperature
- Humidity: 85% — High fungal spore risk (>75%)
- Rain Risk: 0% — Dry foliar spray window
- Wind: 2 km/h — Ideal droplet deposition

SATELLITE + SOIL + WEATHER → TRIANGULATED AGRICULTURAL CONTEXT

High humidity (85%) warns of fungal spread, clay-heavy soil guides irrigation intervals, and NDVI confirms localized rather than plot-wide decline.

Crop & Regenerative Advisory: What Farmers Can Do Next

Moving beyond disease identification: empowering farmers with climate-smart rotations and regenerative soil practices.

RECOMMENDED ROTATION: Cotton / Kapas

- Agro-fit: Thrives in deep black clay-dominant soils of Andhra Pradesh.
- Season match: Current Kharif cycle matches thermal & solar curves.
- Soil benefit: Deep taproot system breaks soil compaction naturally.
- Water requirement: Medium irrigation tolerance fits rainfall forecast.

SUCCESSION CROP: Black Gram / Urad (Legume)

- Nitrogen fixation: Rhizobium nodules naturally replenish soil N.
- Break pest cycles: Non-host crop disrupts Begomovirus & whitefly vectors.
- Soil conditioning: Restores organic carbon (SOC) depleted by heavy feeders.
- Low water requirement: Highly resilient short-duration catch crop.

REGENERATIVE AGRICULTURAL PRACTICES

✓ Crop Residue Mulching

Spreading retained stubble conserves soil moisture, prevents surface crusting, and buffers root temperature in high clay soils.

✓ Biological Bio-Control

Prioritizes 5% Neem seed kernel extract (NSKE) and yellow sticky traps over synthetic organophosphates to protect beneficial pollinators.

✓ Organic Carbon Enrichment

Farmyard manure (FYM) and Jeevamrutha applications to systematically improve the 8.4 g/kg baseline SOC level.

✓ Responsible Framing

Strictly presented as 'potential regenerative practices' — preserving honesty without promising unrealistic guaranteed yields.

Verify Again: Closing the Agricultural Advisory Loop

Most AI agritech tools end at diagnosis. KisanVue AI introduces a continuous verification loop to assess real treatment efficacy.



Chilli Leaf Curl Virus (Begomovirus)

HIGH RISK

Chilli (మిరప / मिर्च)
Baseline diagnosis indicated upward curl and whitefly infestation. Advisory instructed neem oil application and yellow sticky traps.



Quick Test Sample: Recovered Chilli (5-Day Post Advisory)

Days Elapsed Post Advisory: 5 days

Analyze Recovery Progress

STATUS: IMPROVINGAI-Assisted Visual Improvement Score: 85/100POWERED BY GOOGLE GEMINIListen to Advisory

Crop Health Progress: IMPROVING

Based on a visual comparison of the submitted images (not a laboratory measurement), visible indicators suggest improvement. The follow-up plant shows reduced leaf curling and healthy new flower and foliage growth compared to the baseline scan.

* Based on visual comparison of submitted images. Not a laboratory measurement.

PREVIOUS RISK
HIGH

→
Risk Reduced

CURRENT RISK
LOW

Observed Recovery Milestones

- Reduction in leaf curling and upward puckering compared to baseline

Ongoing Field Maintenance

- Continue monitoring vector populations like whiteflies regularly
- Maintain balanced irrigation and nutrient management to support

THE 'VERIFY AGAIN' ENGINE

- The Industry Blind Spot**
Traditional farmer apps give a single prescription and vanish. Farmers never know if treatments worked or wasted money.
- Multimodal Temporal Reasoning**
Google Gemini compares baseline Day-0 foliar symptoms against Day-5 follow-up photographs to evaluate recovery trajectory.
- Quantitative Improvement Score**
Outputs an 85/100 Visual Improvement Score, risk reduction delta (HIGH → LOW), and milestone progression.
- Rigorous Scientific Honesty**
Explicitly labeled: 'AI-assisted visual improvement assessment — not a laboratory measurement'.
- Adaptive Recalibration**
If symptoms worsen, the engine flags non-responsiveness and immediately triggers escalation to a local KVK agronomist.

Built for India: Localized, Multilingual, Digital Public Good

Designed for India's linguistic, regional, and climatic diversity — making complex AI accessible to every marginal farmer.

English

Technical & Global Ag-Tech

Standard agronomic taxonomy, international soil definitions, and RESTful developer APIs.

తెలుగు (Telugu)

Andhra Pradesh & Telangana

చూడండి. అర్థం చేసుకోండి. ఆచరించండి.
ధృవీకరించండి.

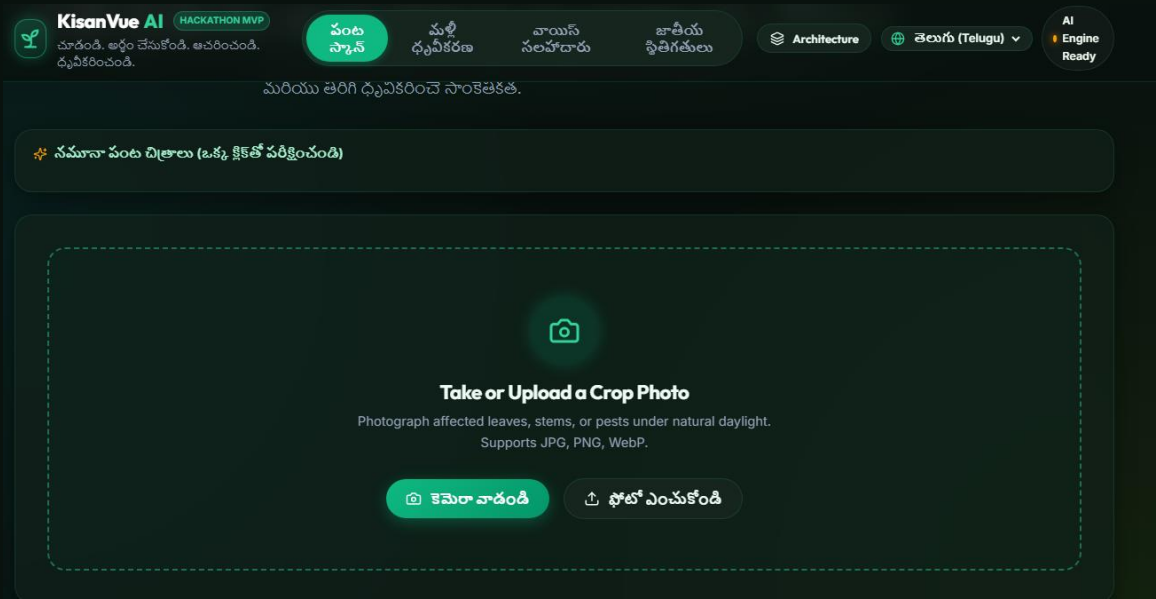
Localized vernacular advisory for chilli, cotton & paddy belts.

हिंदी (Hindi)

Northern & Central Hindi Belt

देखें. समझें. कार्य करें. सत्यापित करें.

Accessible agro-terminology for wheat, mustard, and soybean farmers.

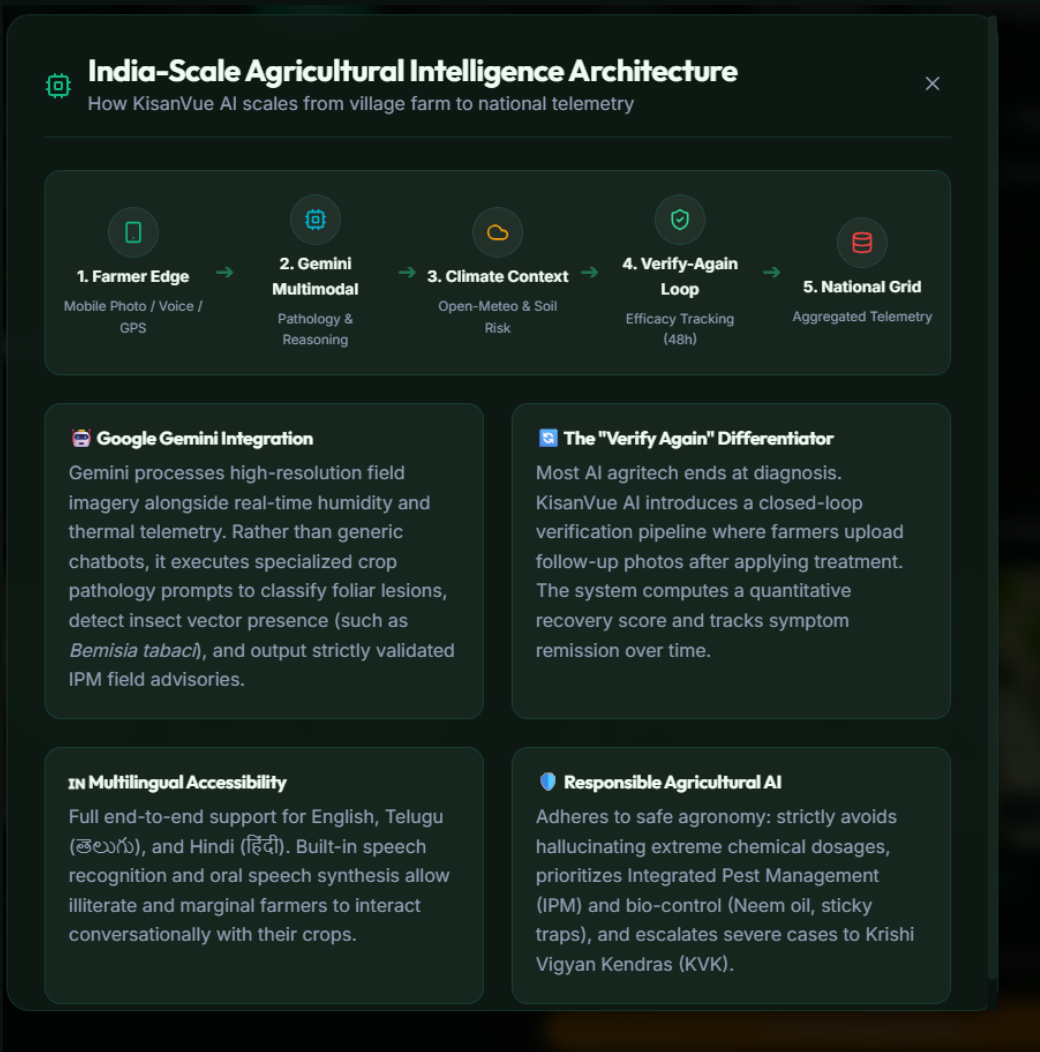


DIGITAL PUBLIC GOOD ARCHITECTURE

- Scalable Hierarchy: Field → Village → Block → District → State level intelligence.
- No App Store Barrier: Zero-install progressive web application (PWA) running on mobile browsers.
- Voice-First Inclusivity: Speech synthesis and voice agronomist for non-literate farmers.
- Public Architecture: Designed to interoperate with open agricultural data standards across Indian agro-climatic zones.

Technology Architecture: Composable Agricultural Intelligence

Production stack built for high reliability, rapid inference, and graceful degradation across variable rural connectivity.



Potential Impact & Scalability: From One Field to National Telemetry

How localized farm insights aggregate into regional agricultural telemetry without compromising individual farm privacy.

1. SINGLE FIELD

Leaf photo + localized micro-climate + SoilGrids physical profile.

2. FARMER

Actionable IPM advisory, vernacular voice guidance & Verify-Again.

3. MANDAL / BLOCK

Early detection of vector hotspots (e.g. Whitefly in Chilli).

4. DISTRICT / STATE

Aggregated crop stress maps informing regional agricultural extension.

5. NATIONAL GRID

Composable, standardized agricultural APIs for research & public good.

PROJECTED VALUE PROPOSITIONS & SCALABILITY PATHWAYS

- Preventing Over-Chemicalization: By prioritizing bio-controls (Neem oil, sticky traps) and soil mulching, farmers avoid excessive chemical expense and soil toxicity.
- Early Stress Interception: Fusing 10m Sentinel-2 NDVI with real-time humidity alerts catches fungal outbreaks days before visual catastrophic crop loss occurs.
- Trust via Closed-Loop Verification: Farmers can track and prove recovery before spending additional capital on follow-up treatments.
- Frugal Rural Deployment: Operates efficiently on mobile browsers without requiring high-end smartphones or expensive proprietary sensors.

"KisanVue doesn't stop at diagnosis."

SEE

Farmer foliar scan & satellite Earth observation.

UNDERSTAND

Google Gemini multimodal agronomic reasoning.

ACT

Localized, soil-aware, weather-timed IPM advisory.

VERIFY

Follow-up visual score closing the loop.

KISANVUE AI — AI-Powered Agricultural Intelligence for India

Live Demo Application: <https://kisanvue-ai.vercel.app/>

GitHub Repository: <https://github.com/HEMANTHSWAMY26/KISANVUE-AI>

Track 4: Agricultural Intelligence